

Linear Electric Machines, Drives and MAGLEVs: an Overview

Ion Boldea, *Life Fellow, IEEE*, Lucian Tutelea, *Member, IEEE*, Wei Xu, *Senior member, IEEE*, and Maracello Pucci, *Senior Member, IEEE*

Abstract— Linear motion is rather common in industry and linear electric motors (LEMs) can provide it directly (without a mechanical transmission), through electromagnetic field forces. LEMs may be considered counterparts of rotary electric machines but specific topologies lead to characteristics that differ (in some cases notably) from those of the latter. The present paper attempts an overview on recent progress of LEMs, from innovative topologies to advanced modeling, design and control, with case studies and examples related to specific industrial applications from people movers to small compressors solenoids, speakers and microphones.

Index Terms— Linear electric motor (LEM), linear induction motor, linear dc excited synchronous motor, linear permanent magnet (PM) synchronous motor, plunger-solenoid linear oscillatory machine, MAGLEV.

I. INTRODUCTION

LINEAR electric (electromagnetic) machines (LEMs) realize the conversion of electrical energy to linear motion mechanical energy (or vice versa) directly through electromagnetic forces. Linear motion is rather common in industry. First LEMs have been invented as early as 19th century but gained industrial attention only after 1960 due to the need of power electronics for motion control (in absence of a mechanical transmission). The present paper attempts an overview on recent progress of LEMs, from innovative topologies to advanced modeling, design and control, with case studies and examples related to specific industrial applications.

Typical industrial applications of LEMs are briefly summarized as follows:

- (1) Magnetically levitated vehicles (MAGLEVs): see

Manuscript received March 27, 2017; revised June 6, 2017; accepted June 28, 2017. Corresponding author: Wei Xu (e-mail: weixu@hust.edu.cn). This work was supported by the National Natural Science Foundation of China under Grants 51377065.

I. Boldea is with the Romanian Academy – Timisoara Branch, Timisoara 300223, Romania, and also with the Politehnica University Timisoara, Timisoara 300006, Romania (e-mail: ion.boldea@upt.ro).

L. Tutelea is with the Department of electrical Engineering, Politehnica University Timisoara, Timisoara 300006, Romania (e-mail: lucian.tutelea@upt.ro).

W. Xu is with the State Key Laboratory of Advanced Electromagnetic Engineering and Technology, School of Electrical and Electronic Engineering, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: weixu@hust.edu.cn).

M. Pucci is with the Institute on Intelligent Systems for Automation – Italian National Research Council, Palermo 90128, Italy (e-mail: pucci@pa.issia.cnr.it).

Shanghai Transrapid,

(2) Urban people movers: see Dallas-Forth Worth’s Airline Commuter,

(3) Vessel rudders, bow thrusters,

(4) Linear oscillatory permanent magnet motor-driven refrigerators/compressors,

(5) Espresso cafe steamer drives,

(6) Digital camera zoomers, clean room industrial transporters,

(7) Linear electric generators for deep space missions or for wave energy conversion,

(8) Hotel door locker solenoids,

(9) Electric power switch solenoids,

(10) Microphones and loudspeakers in cellular phones.

Meanwhile LEMs can be characterized by the following aspects:

(1) Low initial cost (especially in small excursion applications),

(2) High reliability,

(3) Low energy consumption per job done,

(4) Adhesion-free propulsion (lighter vehicles: lower wh/passenger/km),

(5) Better position tracking (no backlash) in industrial positioning.

However, to outperform rotary electric motor drives (with included transmission), LEMs need high performance power electronics control and application adapted topologies and design methodologies. The effort so far in the field is summarized in the quite a few books (available to English readers) dedicated to the subject [1-12]; but, also, in numerous conference papers (many within LDIA triannual conferences) and Journal papers. This paper attempts an overview on recent progress in LEMs. So the paper continues with: LEMs classification in Section II, linear Induction Motors in Section III, linear dc excited synchronous motors in Section IV, Linear PM synchronous motors in Section V, plunger-solenoids in Section VI, linear oscillatory machine in Section VII, x-y motion LEMs in Section VIII, passive guideway attraction levitation system MAGLEV in Section IX. Though the “breadth” of the contents warrants an almost “aerial” qualitative view, strong efforts were made throughout to shed light on quantitative energy conversion and control performance.

II. LEM CLASSIFICATION

LEM classifications may be performed by topology (tubular

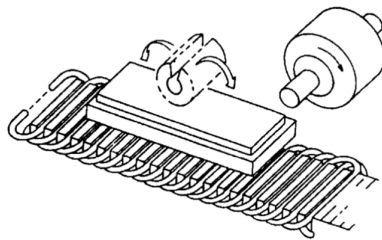


Fig. 1. Typical linear electric machines topology as obtained from a rotary machine.



Fig. 2. Linimo MAGLEV (HSST) in Japan, with single-sided LIM propulsion.

or flat, single or double sided), principles, applications or nature of linear motion: progressive or oscillatory as illustrated synthetically in Fig. 1 [12, 13].

By principle LEMs are categorized by

- (1) Linear induction motors (LIMs) shown in Fig. 2 in a passive guideway MAGLEV,
- (2) Linear dc Excited Synchronous illustrated in Fig. 3 in an active guideway MAGLEV,
- (3) Linear Permanent Magnet (PM) Synchronous motors (LPMSM) shown in Fig. 4 (a),
- (4) Linear brush dc PM motors shown in Fig. 4 (b),
- (5) Solenoids shown in Fig. 5,
- (6) Linear oscillatory PM compressor motors shown in Fig. 6.

The variety in topologies is striking but, in fact, all LEMs may be assimilated as three or single phase counterparts of corresponding rotary machines. However, the solid nature of secondary (aluminum sheet or iron) and the open longitudinal ends of primary magnetic core lead to transverse and skin effects and, respectively, to dynamic longitudinal effects, which are most grave in consequences in LIMs at high speeds (above 10 m/s, in general).

III. LINEAR INDUCTION MOTORS (LIM)

LIMs core has 3 phase windings in uniform slots which in general form a total of $2p$ poles (single layer) or $(2p+1)$ poles (double layer), as shown in Fig. 7.

When $p=1, 2$, there is a phase current asymmetry due to the nonsymmetrical position of the three phases with respect to magnetic core ends, which is called static end effect. For two

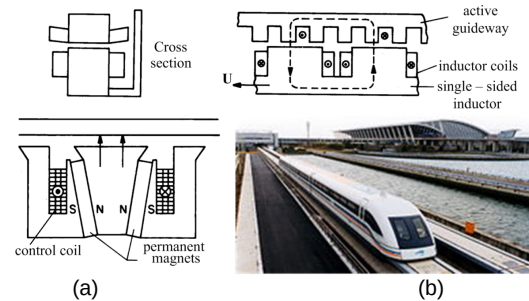


Fig. 3. Active guideway LSM with dc excitation on board of MAGLEV vehicle (Transrapid, Germany): (a) LSM for propulsion and magnetic levitation, (b) The first commercial transrapid.

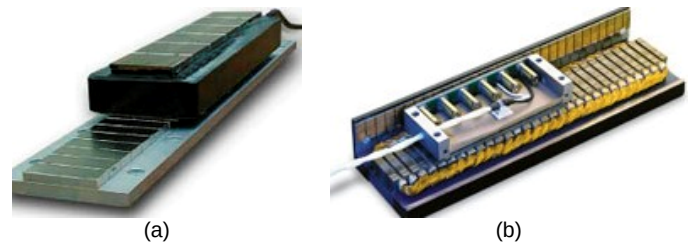


Fig. 4. Commercial linear motors: (a) Commercial iron-core double-sided PM-LSM, (b) Commercial brush-dc PM linear motors.

	Tubular solenoids	Low profile solenoids	Soft shift solenoids	DC Open Frame	AC Open Frame	Laminates
Solenoid power	Average power consumption; moderate force output	Average power consumption; high force output	Average power consumption; moderate force output	Higher power consumption; moderate force output	Higher power consumption; moderate force output	Higher power consumption; highest force output
Solenoid life	25+ million actuations for STA models	1 to 5 million cycles	10 million cycles	50,000 to 100,000 cycles	50,000 to 100,000 cycles	50,000 to 100,000 cycles
Solenoid operation	Push/Pull engagement; well suited to lock/latch operations	Push/Pull engagement; well suited to lock/latch operations	Quiet operation with 3-5 times the starting force of standard solenoids	Pull-in engagement	Pull-in engagement	Pull-in engagement
Force	Up to 97.86 N	Up to 355.86 N	Up to 133.45 N	Up to 133.45 N	Up to 155.7 N	Up to 133.45 N
Stroke	Up to 63.5 mm	Up to 9,525 mm (4.76mm nominal)	Up to 9,525 mm	Up to 25.4 mm	Up to 25.4 mm	Up to 31.75 mm

Fig. 5. Sample commercial solenoids.

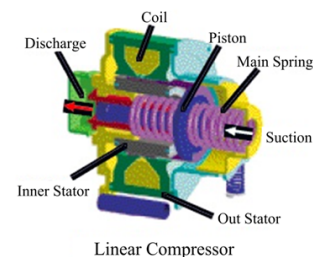


Fig. 6. Commercial linear compressor drive.

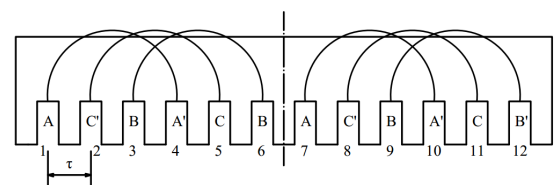


Fig. 7. Single layer three phase LIM windings, $2p=4$.

orthogonal windings such asymmetries are practically eliminated.

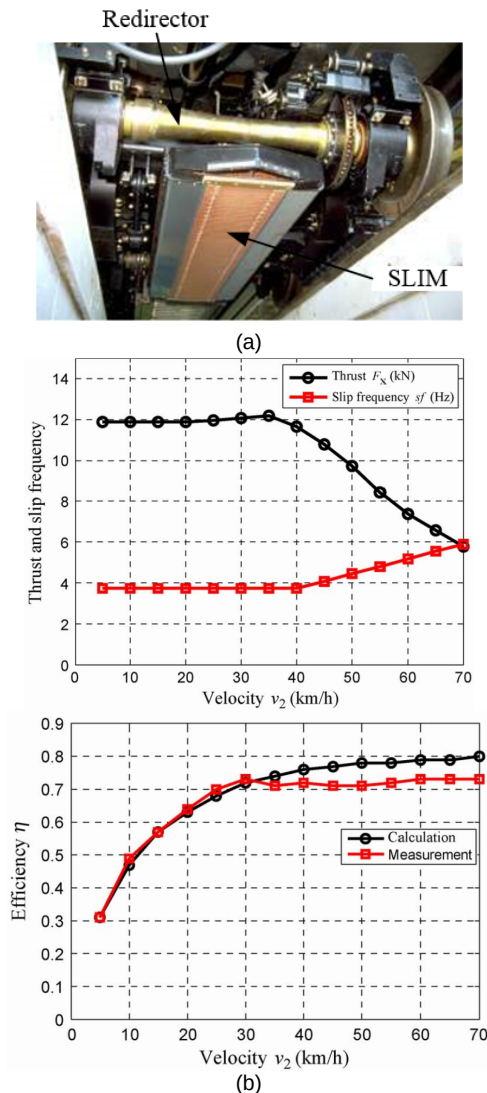


Fig. 8. Performance curves of SLIM based on linear metro: (a) Physical structure, (b) Thrust, slip, power factor and efficiency.

While LIMs specific phenomena mentioned above have been given utmost attention through analytical filed methods, to prove good energy conversion efficiency (above 80%), the power factor remains low, 0.6-0.7, as the ration of airgap to pole pitch τ is not small enough, due to mechanical constraints on minimum air gap value. Some key performance indexes for single-sided LIM (SLIM) based linear metro, such as thrust, efficiency, power factor, *etc.*, are summarized in [14-16], Fig.8 [14]. Recently, 3D FEM coupled field circuit models have been built for LIMs for better modeling at the cost of more than 80 hours of computer time on an 800 MHz PC [17].

Recognizing the loss increase and power factor decrease in LIMs due to dynamic longitudinal effect, the optimal goodness factor concept [3] was introduced even for urban people mover applications (10-30 m/s). PM spinning wheel at primary entry was also proven to produce some net benefit in performance (cost) by reducing the dynamical longitudinal effect.

But the main effort was (is still) directed to simple enough but practical equivalent circuits that account for dynamic

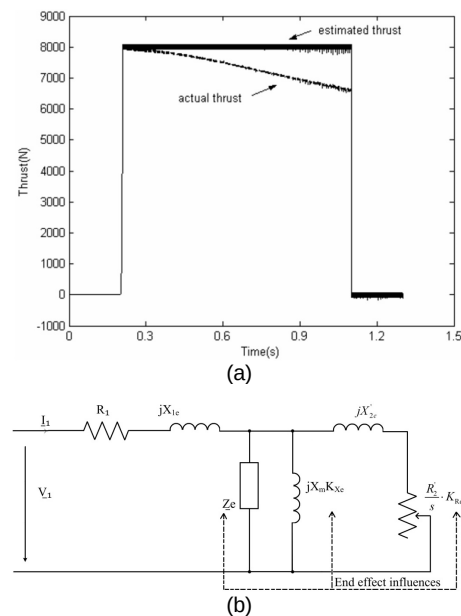


Fig. 9. Dynamic performance of SLIM: (a) Urban SLIM thrust control response: estimated thrust and actual thrust, (b) More complete equivalent circuit of LIM with dynamic longitudinal effect.

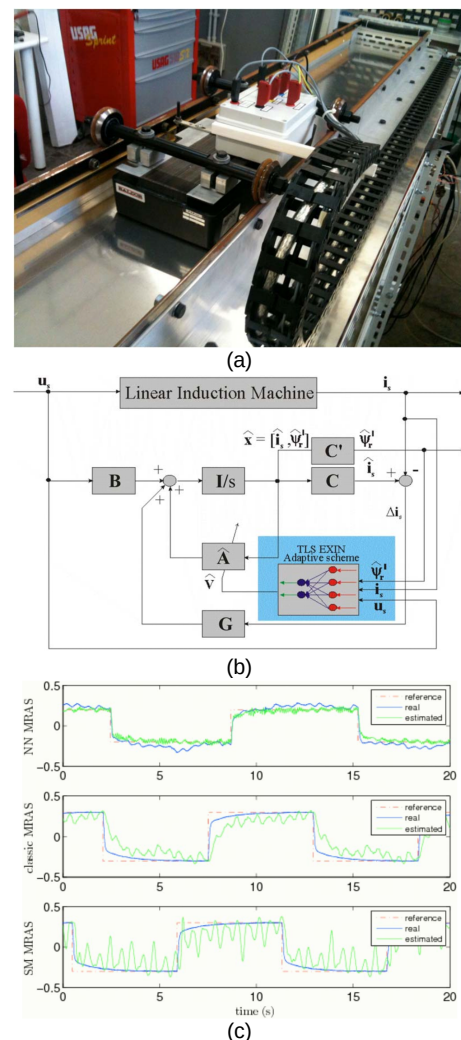


Fig. 10. LIM neural sensorless control with end effects: (a) The 200 N 6.85 m/s, 20 kg, prototype, (b) TLS full order Luenberger observer, (c) Comparative estimated speed of three observers.

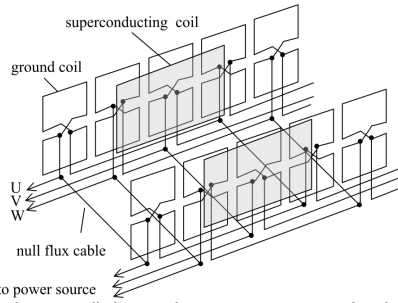


Fig. 11. Eight-shape-coil three-phase ac superconductive winding with propulsion, suspension and guidance force production.

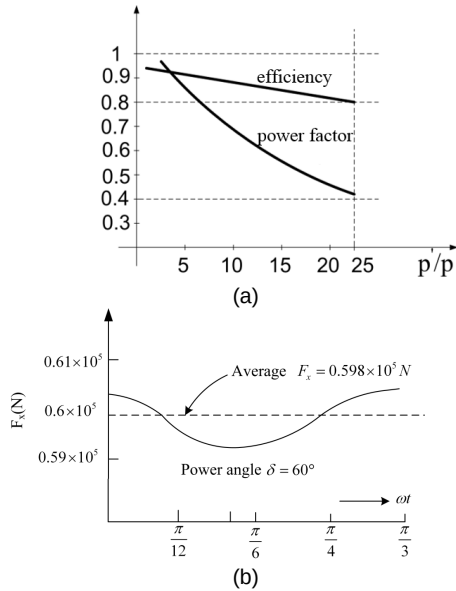


Fig. 12. Performance curves of LSM: (a) Typical efficiency and power factor of a DCE-LSM 45 ton MAGLEV at 110m/s change with the active section length to vehicle length ratio (p'/p), (b) SM-LSM thrust pulsations at 110 m/s.

longitudinal effect and on airgap variations, magnetic saturation, to assist in the vector or direct thrust control system design of medium and large speeds LIMs, as shown in Fig. 9 (a), [18-22].

Even at 10 m/s the dynamic longitudinal end effect coefficient K_e , which appears in Fig. 9(b) [19, 20] has to be accounted for. Advanced control of LIMs with dynamic end effect with motion sensor or sensor-less is illustrated in [21-23], Fig. 10 [22].

The ruggedness of LIMs and their solid secondary, in spite of low power factor (to be compensated by the inverter), but for satisfactory efficiency may become dominant in urban and suburban MAGLEVs, grace to optimal design and advanced (vector and direct thrust) control.

IV.LINEAR DC EXCITED ACTIVE GUIDEWAY SYNCHRONOUS MOTORS

In linear dc excited synchronous motors (DCE-LSM), dc heteropolar excitation on board of mover (with longer (active) three phase primary) is provided at room temperature in iron core machines, as shown in Fig. 3, or by superconducting coils, as shown in Fig. 11 [25]. Both LEMs produce controlled propulsion and levitation but the SC configuration in Fig. 12

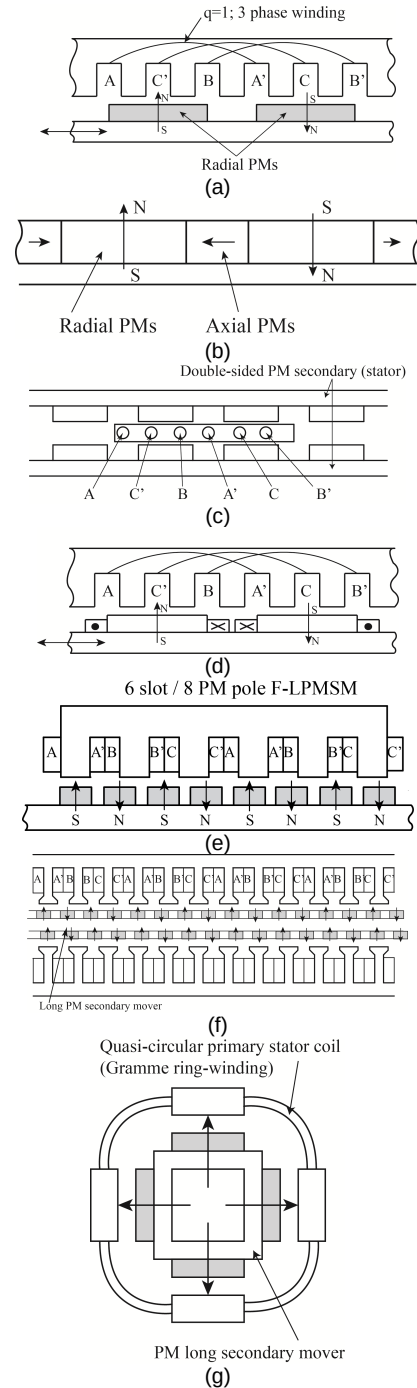


Fig. 13. A few practical flat LPMSMs (F-LPMSMs) topologies: (a) With $q=1$ slot/pole/phase winding primary and radial surface PMs, (b) Radial plus axial (Halbach) PM secondary, (c) Double sided F-LPMSM, (d) With hybrid PM+dc coil secondary for flux weakening or levitation control, (e) Single sided F-LPMSM with tooth-wound three phase winding (6 slot/8 pole), (f) Double sided PM secondary with shifted PMs F-LPMSM (12 slot/16 pole), (g) Four sided configuration with quasi circular coils Gramme winding.

also produces guidance with a more complex active guideway.

The higher initial track cost and political reasons still stop their worldwide application at 400 (500) km/h commercial speeds for intercity travel with lighter vehicles for lower track maintenance. Replacing the superconducting coils with PMs on board has been recently tested for an urban active guideway MAGLEV [26].

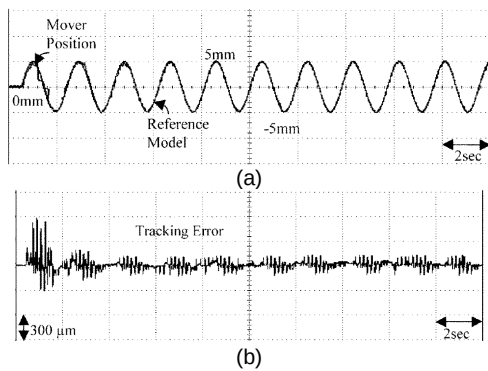


Fig. 14. Tests results with RBMF-based FNN position FOC of F-LPMSM: (a) Tracking response at parameter variation, (b) Tracking error at parameter variation.

V. LINEAR FLAT (AND TUBULAR) PMSMS

Quite a few flat (or tubular), as shown in Fig. 13, LPMSMs have been introduced for industrial (short travel) transport on linear bearings or with controlled (active) magnetic suspension.

A super high thrust flat LPMSM has been shown capable of 30 kN thrust for vessel rudders [27].

The main reason is lower losses per Newton thrust per kg of active materials. Recent progress in this area may be summarized as follows:

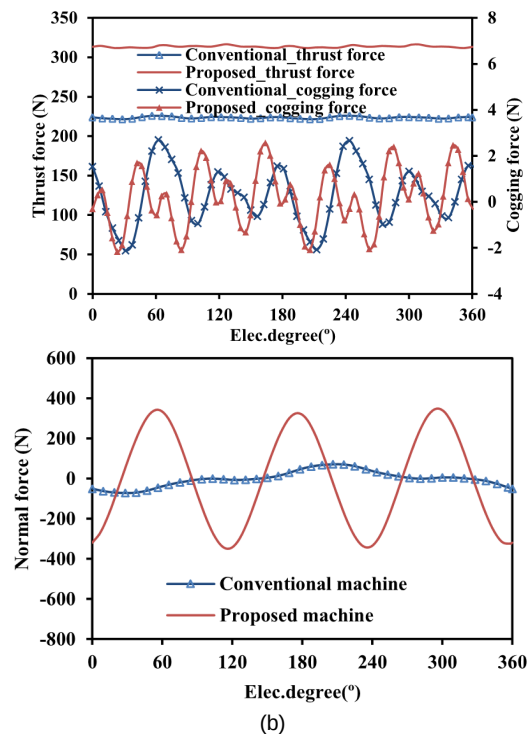
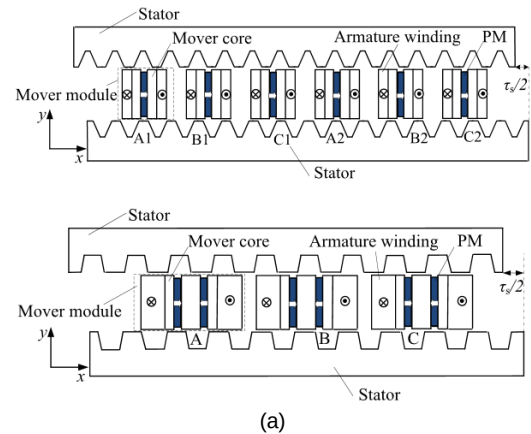
- (1) Advanced magnetic equivalent circuit (MEC) models [28],
- (2) Total cogging force reduction [29],
- (3) Vector [30] and direct thrust control [12] with cogging force compensation, and the test results are shown in Fig 14 [30],
- (4) Advanced analytical field methods for tubular LPMSMs (besides 2D, 3D, FEM investigations), to save some computation time [31].

New topologies such as vernier LPMSMs with auxiliary, dc excitation [32] or magnetically geared tubular LPMSM [33], Fig. 15 has been introduced for particularly low speed high thrust applications (total energy conversion) [34, 35].

VI. FAST ACTION CONTROLLED PLUNGER SOLENOID

Plunger solenoids are required for fast motion response (millisecond range). However, in this case, for the favored symmetric cylindrical structure, magnetic field penetration (even in soft magnetic composite (SMC) cores) becomes a limitation, to be accounted for in complex models, as shown in Fig. 16 [36].

Solenoids may be PM assisted for better dynamic and energy conversion performance (for a valve actuator) and direct FEM geometrical optimization was exercised on them with good results as shown in Fig. 17, in sensor-less position control from 1 to 80 Hz, for ± 3 mm maximum excursion and with mechanical springs [37]. The main idea of the model, as illustrated in Fig. 17 (b), is to derive the electromagnetic force from the mover position and current, and the force generated by the spring is determined by the mover position, then the equation of motion can be obtained and the speed and position



	Proposed	Conventional
Stator height (mm)	36.1	28
Mover length (mm)	199	256
Mover thickness (mm)	100	100
Current density (A/mm ²)	2.35	2.77
Average thrust force (N)	313	224
Thrust density (MN/m ³)	0.449	0.25
Copper loss (W)	34.5	38
Iron loss (W)	7.85	6.76
Total loss (W)	42.35	44.76
Power factor	0.88	0.93
Efficiency	0.88	0.83
Power factor*efficiency	0.774	0.772

(c)

Fig. 15. Flux-switching linear PMSM: (a) Single PM pole modular stator, (b) Thrust and normal force, (c) Yokeless motors performance at 1 m/s.

can be derived.

Mover acceleration levels of 5000 m/s² have been produced experimentally for a maximum of 500 N of thrust, for average copper losses of less than 120 W in a 0.2 liter/actuator volume device. The PMs latch the 30 grams mover in extreme positions (closed and open valve position) to reduce losses. The coils

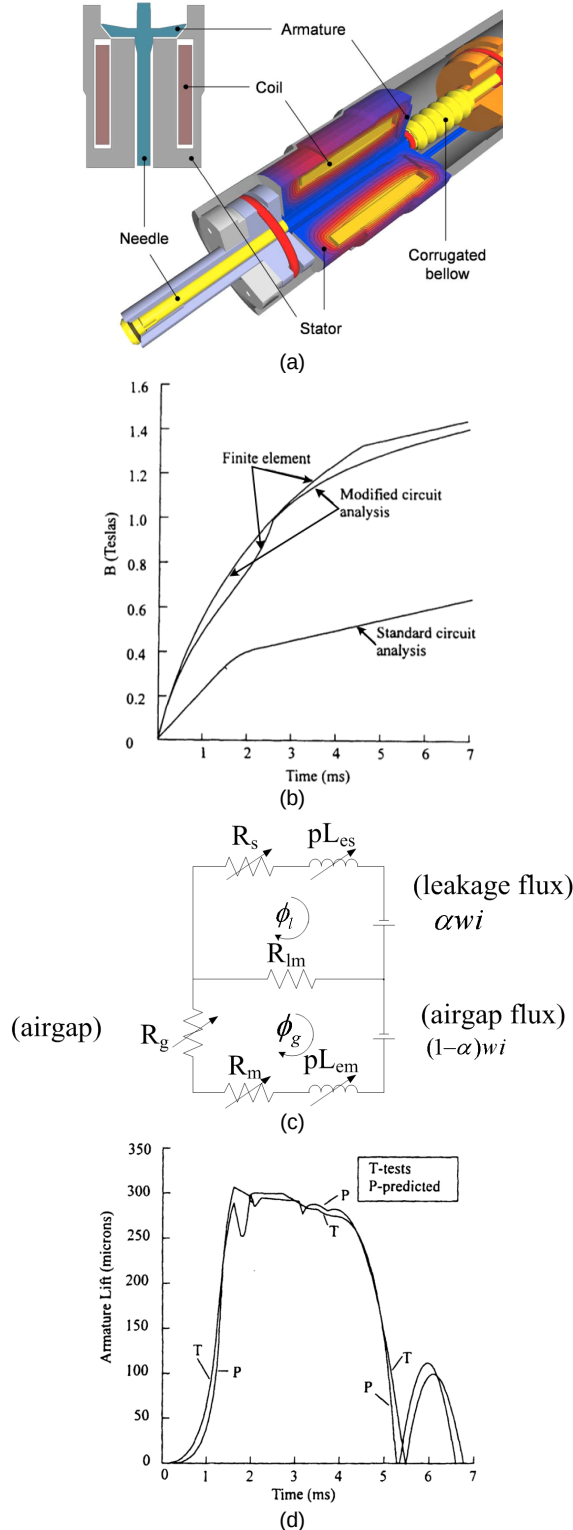


Fig. 16. Plunger solenoids: (a) PM-less solenoid as on-off valve actuator, (b) Transient response for a nonlinear flat plate to a 1592 A/m surface field, (c) Dynamic magnetic circuit of plunger solenoid, (d) Position dynamic transients – tests versus predicted results.

differential action controlled with a single 1 phase inverter allows fast motion dynamics at reasonable total initial cost. This device may be adequate for individual valve actuators in internal combustion engines. Fast action solenoids with PM assistance, as shown in Fig. 18, are applied to power switchless

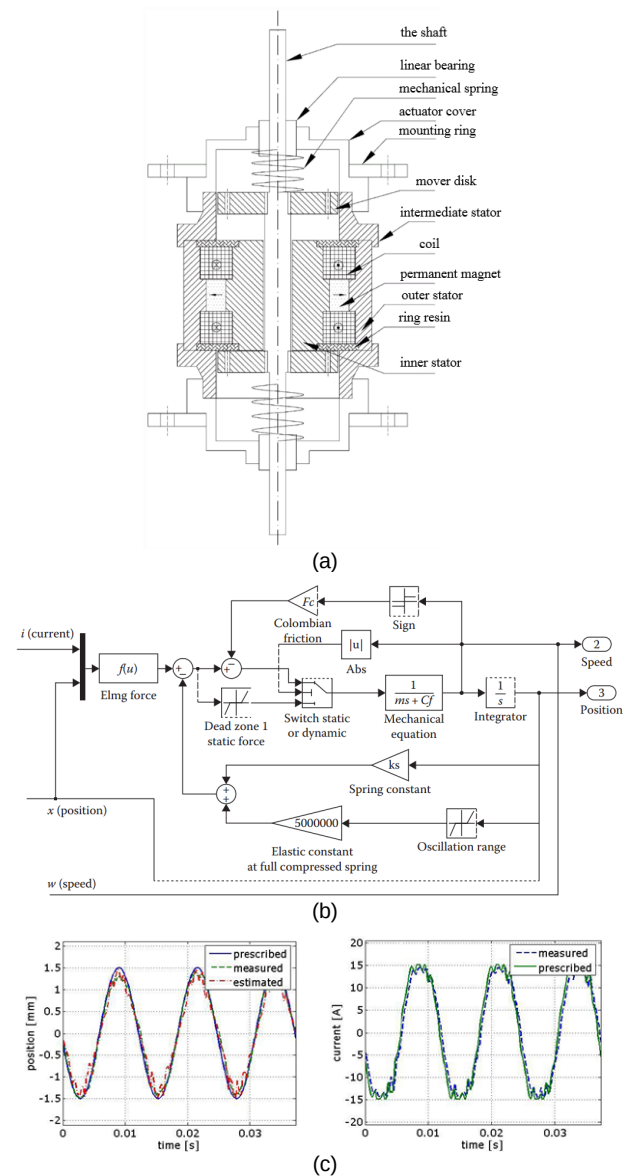


Fig. 17. Valve actuator: (a) Topology of a PM twin coil valve actuator, (b) Mechanical part of actuator model, (c) Close-loop position sensorless control at 80 Hz.

(contactor) [38].

VII. RESONANT OSCILLATORY SINGLE PHASE PM MOTOR/GENERATOR (PM-LOMS)

PM-LOMs are built with coils, or PMs, or iron movers and they operate when electrical current frequency f_e is equal to mechanical eigen frequency f_m of the mover with a mechanical spring constant k_{spring} . Their relation is shown as below,

$$2\pi f_m = \sqrt{\frac{k_{spring}}{m_{mover}}} = 2\pi f_e \quad (1)$$

In this case efficiencies above 70-75% have been obtained even at 25 W of output power. Now the combined microphone and loudspeaker with coil movers and mechanical springs, applied especially in cellular phones (where space is paramount) was investigated by FEM optimization with remarkable recent results, as shown in Fig. 19 [39].

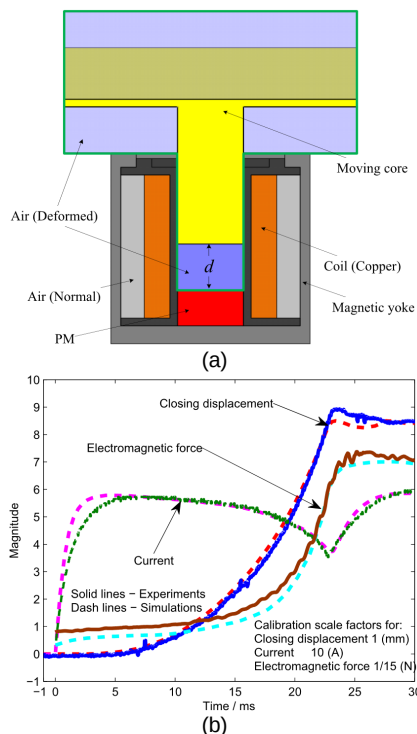


Fig. 18. Magnetic PM contactor: (a) Topology, (b) Current, closing displacement and electromagnetic force for constant voltage (150 Vdc).

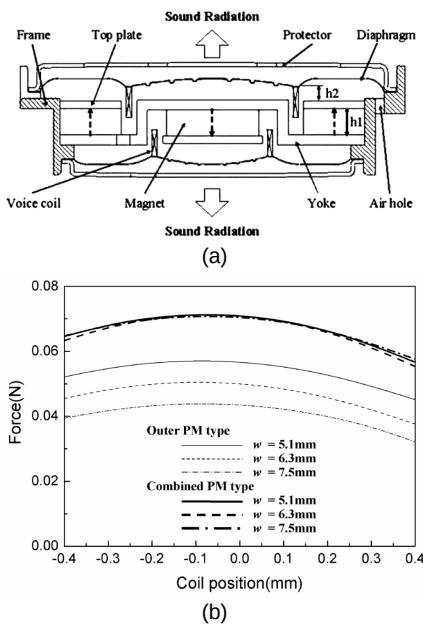


Fig. 19. Combined microphone and speaker for cellular phones: (a) Mechanical part of actuator model, (b) Close-loop position sensorless control at 80 Hz.

Yet another, 25 W, 14 Vdc, such tubular oscillatory motor with ± 3 mm excursion at 270 Hz and outer stator diameter of 16 mm was optimally designed and shown to yield an efficiency of 74% at 60.7 grams total device weight (without the mechanical springs). Only current control was required for stable transients at full power ([12], Chapter 17), because the stroke increases linearly with current after the “pumping point” of a refrigerator reciprocating compressor [31].

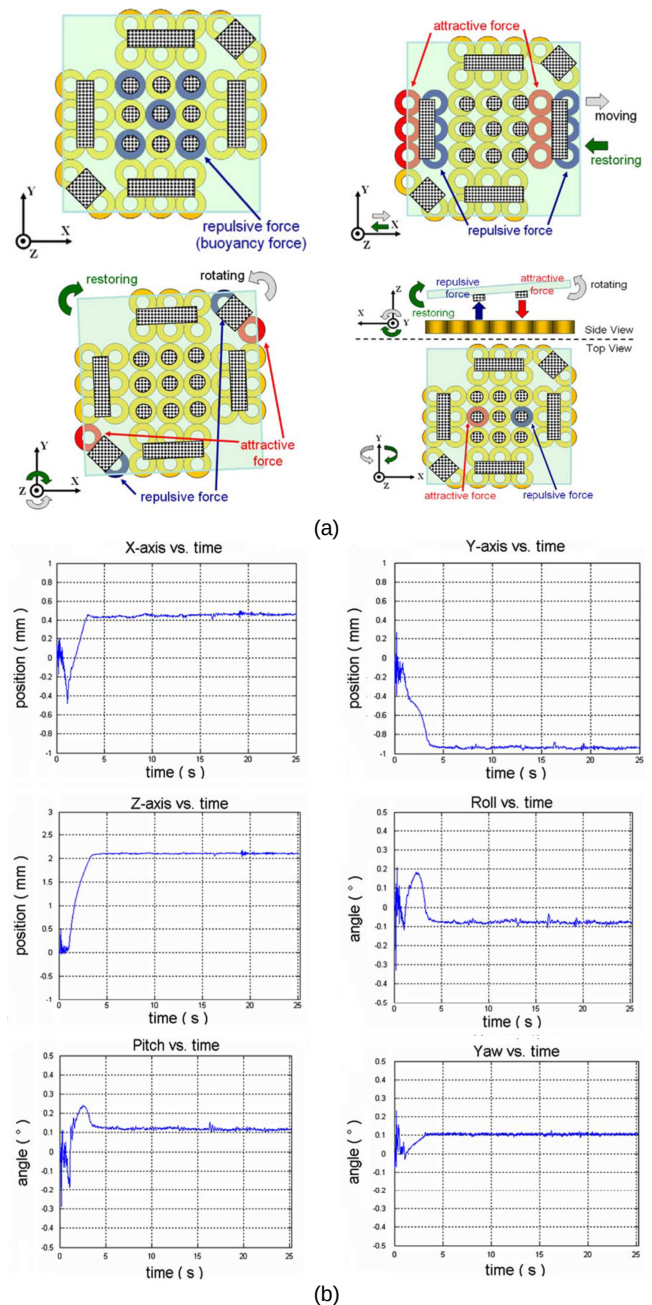


Fig. 20. Six DOF driver of a planar MAGALEV: (a) Six DOF driver of planar MAGLEV stage diagram, (b) Experimental results.

Even a spring-less resonant PM oscillatory motor for 2 kW has been proposed [40] by using the linear to position cogging force as a “magnetic” spring.

VIII. MULTIAXIS LEMS

By multiaxis LEM drives, it means small power devices that produce either long stroke xy (planar) motion with small rotation, without or with active z levitation (vertical) control, or x, y, z (3D) controlled short displacements with precision in the micro- to nanometer range.

Multiaxis LEMs are finding applications in pick and place machines, and with controlled magnetic suspension, which seem adequate for the scope ([12], Chapter 18). The practical dilemma of fixed or moving coils has been recently resolved in

favor of fixed coils and PM-movers. Such a six degree-of-freedom (DOF) driver of a planar MAGALEV is shown in Fig. 20.

The system uses an array of solenoids to levitate and to drive a PM carrier. From the experimental results, it can be found that the response in all six DOF is remarkably stable [41].

IX. ATTRACTION LEVITATION SYSTEMS (ALSS)

Levitation of a body (vehicle) can be controlled by attraction or repulsion magnetic forces. Levitation by “attraction” means magnetic cores with controlled current solenoids while levitation by “repulsion” means dc (superconducting coils or PMs) magnetic field interacting with currents induced by motion in a guideway with fix row, of shorted conductor coils. One works from zero speed (as a “helicopter”) and the other only above a certain take-off speed (“aircraft” style). One is statically unstable the other is statically stable. But both have to be dynamically stabilized by active control; more so ALSs. A typical such ALS is given in Fig. 21 (a) with its linearized system in Fig. 21 (b).

State feedback or sliding mode control is required for good control for wide gap range (as in MAGLEVs, it ranges from 2 to 20 mm) in ALSs; Transrapid also takes the similar control strategies. Control performance without a secondary and tertiary mechanical suspension ($m=5000$ kg) for a large speed MAGLEV shows ALS incapable to provide sufficient ride comfort, as shown in Fig. 22 ([12], Chapter 19). The ALS mass $m=300$ kg means strong mechanical bogie and cabin suspensions, that have to be added when both energy consumption and ride comfort are drastically improved.

In PM assisted ALSs, the so called zero dynamic power (current) sliding mode control may be applied by allowing the airgap to vary with weight, and dynamically, as shown in Fig. 23, unless the airgap becomes dangerously small [42], as shown in Fig. 23 (b), when safety air-gap (non-zero current) control takes over.

This solution fares even better in a passive guideway MAGLEV.

X. PASSIVE GUIDEWAY MAGLEVS

As we already discussed active guideway MAGLEVs a bit, when dealing with dc excited LSMs and with magnetic levitation, as mentioned in Section IX, here we will refer due to the lack of space, only to passive guideway MAGLEVs because of the following points:

- (1) They imply a less expensive guideway.
- (2) They have better energy / mileage correlation.
- (3) They, nowadays, may be extended because low weight high efficiency (IGBT SiC) power electronics and safe mechanical energy transfer to vehicle (up to 500 km/h) are available.

A typical such vehicle with linear LIMs propulsion and dc controlled electromagnetic levitation (ALSs) is shown in Fig. 24.

In an effort to increase overall energy efficiency, a solid iron segmented passive guideway with on board linear homopolar

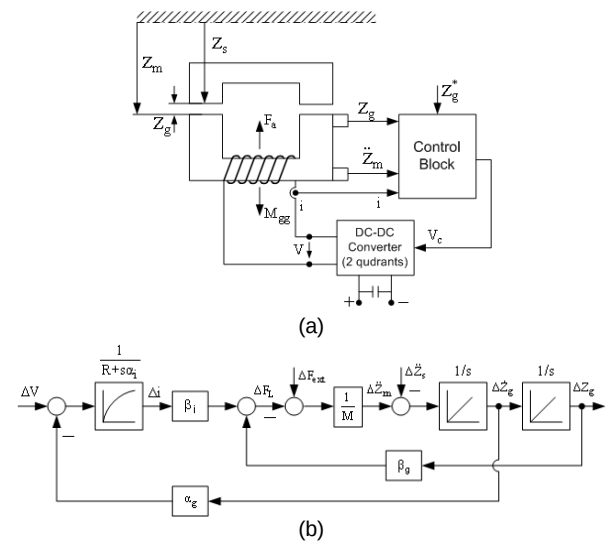


Fig. 21. Typical ALS: (a) Single electromagnet-guideway system, (b) Linearized system with Δi as electrical variable.

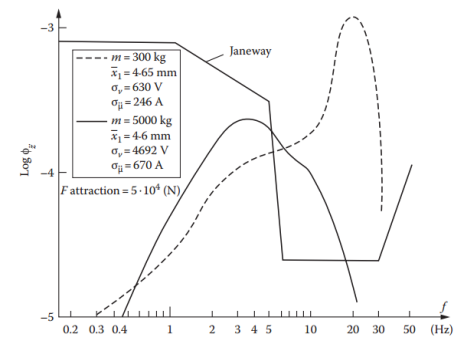


Fig. 22. Ride comfort $m=300$ kg, $M=5000$ kg.

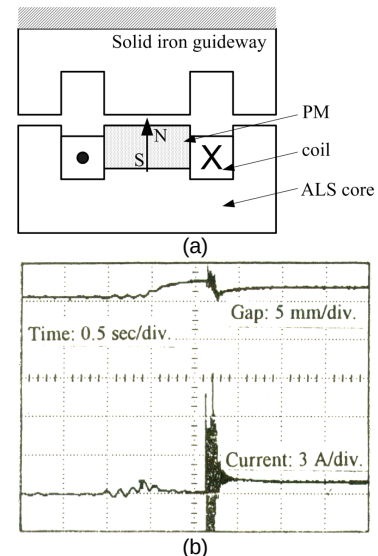


Fig. 23. PM-assisted ALS: (a) PM-assisted ALS diagram, (b) Overload operation (25 kg payload) operation at minimum air-gap (5 mm).

inductor synchronous motor propulsion and levitation vehicle (MAGNIBUS) was proposed, as shown in Fig. 25 [43].

Its takes off and landing dc excitation coils voltage, current and airgap are shown in Fig. 26. The total efficiency/propulsion

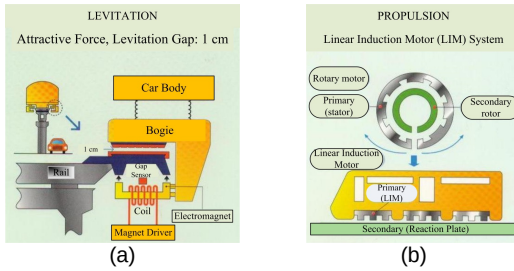


Fig. 24. S. Korean LIM-MAGLEV: (a) Cross-section, (b) Longitudinal view.



Fig. 25. H-LSM based on MAGLEV: (a) Photograph of Magnibus-01, (b) H-LSM MAGLEV system.

TABLE I
MAGNIBUS-01 DATA

Weight	4 ton	passive guideway: solid iron segments $0.25 \text{ m} \times 0.1 \text{ m} \times 0.03 \text{ m}$
Length	4 m	power collectors: 200 V dc (trolleybus double-collectors)
Number of linear inductor motors	4	$3 \times 220/380 \text{ V ac}$ (brush collectors) - to be eliminated
Motor weight	140 kg	levitation force: 10 kN
Number of poles per motors	6	
Rated airgap	0.01 m	test track length: 150 m
Active stack width	0.11 m	levitation: attraction
		- dc controlled
Pole pitch	0.1 m	propulsion: position
		ac controlled

and levitation at 10 m/s was 0.8 and so was the power factor, both notably better than in LIM-MAGLEVs. The 4-tonnes Magnibus-01 prototype main data is given in Table I.

Other practical iron-passive, but laminated, guideway MAGLEVs are still being proposed ([12], Chapter 22), as shown in Fig. 27.

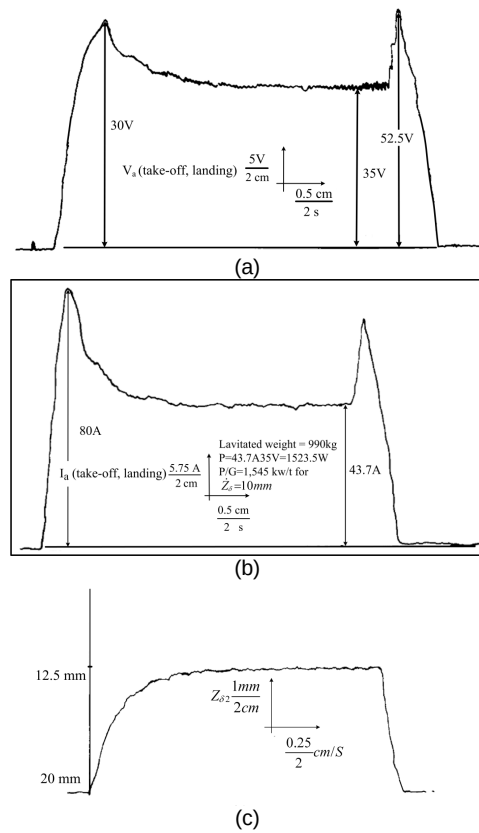


Fig. 26. Magnibus-01: experimental levitation take-off (lifting) and landing at zero speed: (a) Voltage, (b) Dc current and (c) Airgap.

XI. CONCLUSION

Linear motion control, by the direct electromagnetic force of LEMs, is common in industry, from transportation to microphones / loudspeakers and refrigerator compressor drives.

Though invented in the nineteen's century, they have been revived after 1960 by the development of power electronics and of analytical and numerical optimal design tools.

LEMs may be considered as counterparts of rotary electric machines but topological peculiarities (especially the magnetic cores openness in the direction of motion) set them apart.

LIMs above 10 m/s speed have already shown dynamic longitudinal end effects that reduce efficiency and power factor. Advanced analytical and numerical (3D FEM) models are required for design at optimal goodness factor to secure acceptable performance (efficiency around 0.86) and lagging power factor around 0.6-0.65; advanced equivalent circuits accounting for dynamic longitudinal end effects have been recently proven to increase dynamic performance in high speed LIMs that are preferred because of their ruggedness and reasonable cost.

Active guideway linear synchronous motors (LSMs) with dc excitation and magnetic core (or with superconducting dc excitation and "air core") have been applied up to 400 (500) km/h in MAGLEVs; integrated propulsion and levitation over a wide airgap range has been proven with advanced robust (sliding mode) control. LSMs with PM assisted dc excitation

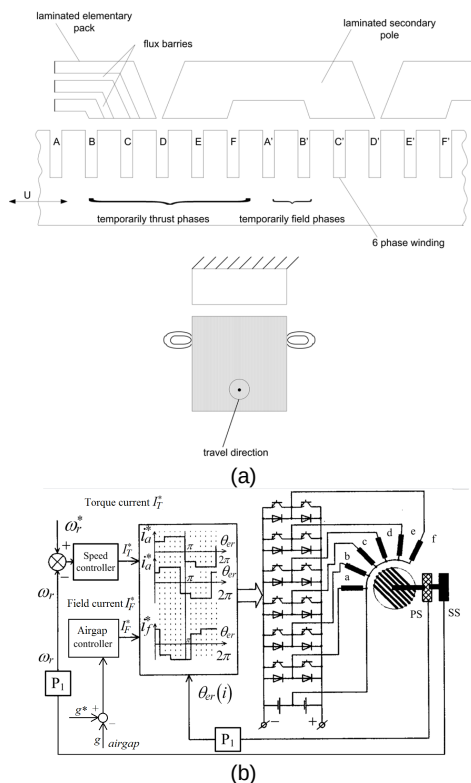


Fig. 27. Generic bipolar two-level current BLDC-MRM MAGLEV: (a) longitudinal cross-section with 6 phase and (b) Inverter dedicated topology.

endowed with zero average power (current) levitation control (control coils), are considered suitable for industrial MAGLEVs along short travels.

Magnetically geared LPMsMs or lead screws [44, 45] have been introduced for low speed high thrust to dramatically increase thrust density for active suspension on vehicles or wave energy conversion.

Fast (millisecond range) action controlled solenoids, without or with PM assistance, are currently used for door lockers, power switches or as valve actuators, developing mover accelerations of up to 5 km/s^2 for combined microphone-loudspeakers in cellular phones, (optimally designed by FEM, to leave more area / volume for the screen). Resonant linear oscillatory machines (single phase with PMs) use a resonating mechanical spring ($f_e=f_m$) and are capable of 74% efficiency already at 25 W output in a ($\pm 3 \text{ mm}$ stroke at 270 Hz) 60 grams device.

The mechanical spring role may be played by a linear to position cogging force, resulted from a special design, without harming the efficiency but reducing the actuator (generator) volume (cost).

Attraction levitation systems can provide controlled magnetic suspension for MAGLEVs in high or medium speed or industrial transport at $10 \pm 2 \text{ mm}$ airgap, 3 kVA / tone peak suspension power at speed up to 400 km/h, provided additional, bogie and cabin, mechanical suspensions are added. Vertical axis large power Darius wind turbines might also be levitated this way economically.

Besides active guideway MAGLEVs, burdened by

complicated (costly) guideway and on-land power electronics stations, passive guideways with on board power transfer and equipment MAGLEVs with linear induction motors propulsion or with linear homopolar inductor synchronous motors integrated propulsion-levitation – MAGNIBUS system – have been vigorously developed for urban people movers or for industrial transport at notably lower track (and system) initial cost. The ever more environmentally and energy conscience society of today and of near future should find LEMs technology more and more suitable for linear motion/energy control.

REFERENCES

- [1] E. Laithwaile, *Induction Machines for Special Purposes*. New York: Newness, 1966.
- [2] S. Yamamura, *Theory of Linear Induction Motors*. New York: John Wiley & Sons, 1972.
- [3] S. Nasar, I. Boldea, *Linear Motion Electric Machines*. New York: John Wiley and Sons, 1976.
- [4] P. Budig, A.C. *Linear Motors*. Berlin, Germany: Veb Verlag Technik, 1978.
- [5] M. Poloujadoff, *The Theory of Linear Induction Machinery*. Oxford, U.K.: Clarendon, 1980.
- [6] I. Boldea, S. Nasar, *Linear Motion Electromagnetic Systems*. New York: John Wiley & Sons, 1985.
- [7] S. Nasar, I. Boldea, *Linear Electric Motors: Theory, Design, and Practical Applications*. Englewood Cliffs, NJ: Prentice-Hall, 1987.
- [8] J. F. Gieras, *Linear Induction Drives*. Oxford, U.K.: Clarendon, 1994.
- [9] I. Boldea and S. Nasar, *Linear Electric Actuators and Generators*. London, U.K.: Cambridge Univ. Press, 1997.
- [10] J. F. Gieras & Z.J. Piech, *Linear Synchronous Motors*. Boca Raton, Florida: CRC Press, 2000.
- [11] I. Boldea, S. Nasar, *Linear Motion Electromagnetic Devices*. New York: Taylor and Francis, 2001.
- [12] I. Boldea, *Linear Electric Machines, Drives and MAGLEVs Handbook*. CRC Press, 2013.
- [13] C. Pompermaier, F. Kalluf, A. Zambonetti, M. Ferreira, and I. Boldea, "Small linear PM oscillatory motor: Magnetic circuit modeling corrected by axisymmetric 2-D FEM and experimental characterization," *IEEE Trans. Ind. Electron.*, vol. 59, no. 3, pp. 1389–1396, Mar. 2012.
- [14] W. Xu, J. Zhu, Y. Zhang, Z. Li, Y. Li, Yi Wang, Y. Guo, and Y. Li, "Equivalent circuits for single-sided linear induction motors," *IEEE Trans. Ind. Appl.*, vol. 49, no. 6, pp. 2410–2423, Nov./Dec. 2010.
- [15] W. Xu, J. Zhu, Y. Zhang, Y. Li, Y. Wang, and Y. Guo, "An improved equivalent circuit model of a single-sided linear induction motor," *IEEE Trans. Veh. Technol.*, vol. 59, no. 5, pp. 2277–2289, Jun. 2010.
- [16] W. Xu, G. Sun, L. Wen, W. Wu, and P. Chu, "Equivalent circuit derivation and performance analysis of a single-sided linear induction motor based on the winding function theory," *IEEE Trans. Veh. Technol.*, vol. 61, no. 4, pp. 1515–1525, May 2012.
- [17] Yamaguchi Tadashi, Kawase Yoshihiro, Yoshida Makoto, Nagai Masato, Ohdachi Yasuharu, and Saito Youichi, "3D FE analysis of a LIM with two armatures," *Record of LDIA 2001*, Nagano, Japan, pp. 300 – 303.
- [18] Fuji Nabuo, Sakamoto Yasuaki, and Kayasuga Takeshi, "End effect compensator based on new concept for LIM," *Record of LDIA 2001*, Nagano, Japan, pp. 102–106.

- [19] I. Takahashi, Y. Ide, "Decoupling thrust and attractive forces of LIM using space vector control inverter," *IEEE Trans. Ind. Appl.*, vol. 29, No. 1, pp. 161 – 167, Jan/Feb 1993.
- [20] C.M.A.R. Cabral, "Analysis of LIMs using new longitudinal end effect factors," *Record of LDIA 2003*, Birmingham, UK, pp. 291 – 294.
- [21] F. Alonge, M. Cirrincione, M. Pucci, and A. Sferlazza, "Input-output feedback linearization control with on-line mars-based inductor resistance estimation of linear induction motors including the dynamic end effects," *IEEE Trans. Ind. Appl.*, vol. 52, No. 1, pp. 254 – 266, Jan.-Feb. 2016.
- [22] A. Accetta, M. Cirrincione, M. Pucci, and G. Vitale, "Neural sensorless control of linear induction motors by a full-order luenberger observer considering the end effects," *IEEE Trans. Ind. Appl.*, vol. 50, No. 3, pp. 1891 – 1904, May-June 2014.
- [23] P. Canelli, V. Delli Colli, F. Marignetti, and I. Boldea, "Sliding mode speed sensorless control of LIM drives," *Record of LDIA 2003*, Birmingham, 8 - 10 September 2003.
- [24] K. Wang, L. Shi, and X. Li, "Direct thrust control of LIM longitudinal end effects," *Record of LDIA 2007*.
- [25] M. Toshiaki, F. Shunsuke, "Design of coil specifications in EDS Maglev using an optimization program," *Record of LDIA 1998*, Tokyo, Japan, pp. 343-346
- [26] B.S. Borowy, H. Guroi, and J-K Kim, "Controller design of linear synchronous motor for general atomics urban MAGLEV," *Record of LDIA 2007*.
- [27] A. Tassarolo, C. Bruzzese, "Computationally efficient thermal analysis of a low-speed high-thrust linear electric actuator with a three-dimensional thermal network approach," *IEEE Trans. Ind. Electron.*, vol. 62, No. 3, pp. 1410 – 1420, March 2015.
- [28] B. Sheikh-Ghalavand, S. Vaez-Zadeh, and A. Isfahani, "An improved magnetic equivalent circuit model for iron-core linear permanent magnet synchronous motors," *IEEE Trans. Magn.*, vol. 46, no.1, pp. 112120, Jan. 2010.
- [29] N. Baatar, H. Yoon, M. Pham, P. Shin, and C. Koh, "Shape optimal design of a 9 slot 10 pole PM LSM for detent force reduction using adaptive response surface method," *IEEE Trans. Magn.*, vol. 45, no. 10, pp. 4562-4565, Oct. 2009.
- [30] L. Zeng, X. Chen, X. Luo, and X. Li, "A vector control method of LPMBDCM considering effects of PM flux linkage and cogging force," *IEEE Trans. Magn.*, vol. 46, no.3, pp. 954-959, March 2010.
- [31] J. Wang, D. Howe, "Tubular modular PM machines equipped with quasi-halbach magnetized magnets- Part I+II," *IEEE Trans. Magn.*, vol. 41, no.9, pp. 2470-2489, Sept. 2005.
- [32] T. Ching, K. Chau, and W. Li, "Power factor improvement of a linear vernier permanent-magnet machine using auxiliary dc field excitation," *IEEE Trans. Magn.*, vol. 52, No. 7, Article Sequence Number: 8204804, July 2016.
- [33] S. Ho, Q. Wang, S. Niu, and W. Fu, "A novel magnetic-gear tubular linear machine with halbach permanent-magnet arrays for tidal energy conversion," *IEEE Trans. Magn.*, vol. 51, No. 11, Article Sequence Number: 8113604, Nov. 2015.
- [34] L. Cappelli, F. Marignetti, G. Mattiazzo, E. Giorcelli, G. Bracco, S. Carbone, and C. Attiaianese, "Linear tubular permanent-magnet generators for the inertial sea wave energy converter," *IEEE Trans. Ind. Appl.*, vol. 50, No. 3, pp. 1817 – 1828, May-June 2014.
- [35] B. Zhang, M. Cheng, J. Wang, and Sa Zhu, "Optimization and analysis of a yokeless linear flux-switching permanent magnet machine with high thrust density," *IEEE Trans. Magn.*, vol. 51, No. 11, Article Sequence Number: 8204804, Nov. 2015.
- [36] J. Caridies, "Fast acting electromagnetic actuators," Ph.D. Thesis, 1982, Dept. of Mech. Eng, Pennsylvania State University, USA.
- [37] S. Agarlita I. Boldea, F. Marignetti, and L. Tutelea, "Position sensorless control of a linear interior PM oscillatory machine, with experiments," *Record of OPTIM 2010*, pp. 689-695.
- [38] L. Shu, L. Wu, G. Wu, and Z. Wu, "A fully coupled framework of predicting the dynamic characteristics of permanent magnet contactor," *IEEE Trans. Magn.*, vol. 52, No. 8, Article Sequence Number: 800160, Aug. 2016.
- [39] S. Hwang, K. Hong, H. Lee, J. Kim, and S. Jeung, "Reduction of harmonic distortion in dual magnetic type microspeakers," *IEEE Trans. Magn.*, vol. 40, no 4, pp. 3054-3056, July 2004.
- [40] I. Boldea, S. Agarlita, and L. Tutelea, "Springless resonant linear PM oscillomotors," *Record LDIA 2011*, Eindhoven, Netherlands.
- [41] Y. Lai; Y. Lee, and J. Yen, "Design and servo control of a single-deck planar maglev stage," *IEEE Trans. Magn.*, vol. 43, No. 6, pp. 2600 – 2602, June 2007.
- [42] Y. Tzeng, and T. Wang, "Dynamic analysis of the MAGLEV system using controlled - PM electromagnets with robust zero power control strategy," *Record of IEEE InterMag*, 1995, San Antonio, Texas, USA.
- [43] I. Boldea, A. Trica, G. Papusoiu, and S. Nasar, "Field texts on a MAGLEV with passive guideway linear inductor motor transportation system," *IEEE Trans. Veh. Technol.*, vol. 37, No 1, pp. 213-219, Nov. 1988.
- [44] N. Berg, R. Holm, and P. Rasmussen, "A novel magnetic lead screw active suspension system for vehicles," *Record of IEEE ECCE 2014*, pp. 3139 - 3146.
- [45] R. Holm, N. Berg, M. Walkusch. P. Rasmussen, and R. Hansen, "Design of a magnetic lead screw for wave energy conversion," *IEEE Trans. Ind. Appl.*, vol. 49, No. 6, pp. 2699 – 2708, Nov.-Dec. 2013.



Ion Boldea (M'77-SM'81-F'96) received the M.S. and Ph.D. degrees in electrical engineering from the Politehnica University of Timisoara, Timisoara, Romania, in 1967 and in 1973, respectively.

He is currently a Full Professor with the Politehnica University of Timisoara. He worked and published extensively in linear and rotary electric machines, drives, and magnetic levitation vehicles. He has published numerous books, with 6000 entrances in libraries worldwide.

Dr. Boldea is a member of the Industrial Drives Committee and the Electric Machines Committee of the OPTIM International Conferences (IEEE-IAS sponsored) from 1996 to present. He received the 2015 IEEE Nikola Tesla Award.



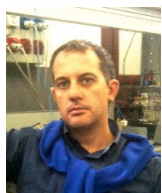
Lucian Nicolae Tutelea (M'07) received the B.S. and Ph.D. degrees in electrical engineering from the Politehnica University of Timisoara, Timisoara, Romania, in 1989 and in 1997, respectively.

He is currently a Professor with the Department of Electric Engineering, Politehnica University of Timisoara. His main research interests include design, modeling, and control of electric machines and drives.



Wei Xu (M'09-SM'13) received the double B.E. and M.E. degrees from Tianjin University, China, in 2002 and 2005, and the Ph.D. from the Institute of Electrical Engineering, Chinese Academy of Sciences, in 2008, respectively, all in electrical engineering.

From 2008 to 2012, he held several academic positions in both Australian and Japanese universities and companies. Since 2013, he has been full professor with the State Key Laboratory of Advanced Electromagnetic Engineering and Technology, Huazhong University of Science and Technology, China. His research topics mainly cover electromagnetic design and control algorithm of linear/rotary machines.



Marcello Pucci (M'03-SM'11) received the Laurea and Ph.D. degrees in Electrical Engineering from the University of Palermo, Italy, in 1997 and 2002, respectively.

In 2000 he has been a host student at the Institute of Automatic Control, Technical University of Braunschweig, Germany, working in the field of control of AC machines, with a grant from the German Academic Exchange Service. Since 2001, he has been a researcher at the Institute of Intelligent Systems for Automation, Section of Palermo, Italy, where he is currently a Senior Researcher.

Dr. Pucci is an associate editor of the IEEE Transactions on Industrial Electronics and IEEE Transactions on Industry Applications. He is a member of the Editorial Board of the "Journal of Electrical Systems".